tabulate twoway — Two-way table of frequencies

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Description

`tabulate` produces a two-way table of frequency counts, along with various measures of association, including the common Pearson's χ^2, the likelihood-ratio χ^2, Cramér's V, Fisher's exact test, Goodman and Kruskal's gamma, and Kendall's τ_b.

Line size is respected. That is, if you resize the Results window before running `tabulate`, the resulting two-way tabulation will take advantage of the available horizontal space. Stata for Unix(console) users can instead use the `set linesize` command to take advantage of this feature.

`tab2` produces all possible two-way tabulations of the variables specified in *varlist*.

`tabi` displays the $r \times c$ table, using the values specified; rows are separated by '\'. If no options are specified, it is as if `exact` were specified for a 2×2 table and `chi2` were specified otherwise. See **[U] 19 Immediate commands** for a general description of immediate commands. See *Tables with immediate data* below for examples using `tabi`.

See [R] **tabulate oneway** if you want a one-way table of frequencies. See [R] **table** and [R] **tabstat** if you want one-, two-, or n-way table of frequencies and a wide variety of summary statistics. See [R] **tabulate, summarize()** for a description of `tabulate` with the `summarize()` option; it produces a table (breakdowns) of means and standard deviations. `table` is better than `tabulate, summarize()`, but `tabulate, summarize()` is faster. See [R] **Epitab** for a 2×2 table with statistics of interest to epidemiologists.

Quick start

Two-way table of frequencies for `v1` and `v2`

```
tabulate v1 v2
```

Add row percentages

```
tabulate v1 v2, row
```

Frequencies only for observations where `v3` $= 1$

```
tabulate v1 v2 if v3==1
```

Weighted cell counts using frequency weights defined by `wvar`

```
tabulate v1 v2 [fweight=wvar]
```

Pearson's χ^2 test and each cell's contribution

```
tabulate v1 v2, chi2 cchi2
```

All available measures of association

```
tabulate v1 v2, all
```

All possible two-way tables for v1, v2, and v3

```
tab2 v1 v2 v3
```

Input cell frequencies and perform χ^2 test

```
tabi 30 18 38 \ 13 7 22, chi2
```

Menu

tabulate

Statistics > Summaries, tables, and tests > Frequency tables > Two-way table with measures of association

tab2

Statistics > Summaries, tables, and tests > Frequency tables > All possible two-way tables

tabi

Statistics > Summaries, tables, and tests > Frequency tables > Table calculator

Syntax

Two-way table

```
tabulate varname1 varname2 [if] [in] [weight] [, options advanced_options]
```

Two-way table for all possible combinations—a convenience tool

```
tab2 varlist [if] [in] [weight] [, options]
```

Immediate form of two-way tabulations

```
tabi #11 #12 [...] \ #21 #22 [...] [\ ...] [, options advanced_options]
```

options	Description
Main	
chi2	report Pearson's χ^2
exact[(#)]	report Fisher's exact test
gamma	report Goodman and Kruskal's gamma
lrchi2	report likelihood-ratio χ^2
taub	report Kendall's τ_b
V	report Cramér's V
cchi2	report Pearson's χ^2 in each cell
column	report relative frequency within its column of each cell
row	report relative frequency within its row of each cell
clrchi2	report likelihood-ratio χ^2 in each cell
cell	report the relative frequency of each cell
expected	report expected frequency in each cell
nofreq	do not display frequencies
rowsort	list rows in order of observed frequency
colsort	list columns in order of observed frequency
missing	treat missing values like other values
wrap	do not wrap wide tables
[no]key	report/suppress cell contents key
nolabel	display numeric codes rather than value labels
nolog	do not display enumeration log for Fisher's exact test
* firstonly	show only tables that include the first variable in *varlist*
Collect	
collect	post results to collection Tabulate
collect([*cname*][, *collect_options*])	post results to a named collection
all	equivalent to specifying chi2 lrchi2 V gamma taub

*firstonly is available only for tab2.

by is allowed with tabulate and tab2; see **[U] 11.1.10 Prefix commands**.

fweights, aweights, and iweights are allowed by tabulate. fweights are allowed by tab2. See **[U] 11.1.6 weight**.

all does not appear in the dialog box.

collect_options	Description
`append`	append results to an existing collection
`replace`	replace results of an existing collection
`label(`*filename*`)`	specify the collection labels
`style(`*filename*`[ , override ])`	specify the collection style

advanced_options	Description
Advanced	
`matcell(`*matname*`)`	save frequencies in *matname*
`matrow(`*matname*`)`	save unique values of *varname*$_1$ in *matname*
`matcol(`*matname*`)`	save unique values of *varname*$_2$ in *matname*
‡ `replace`	replace current data with given cell frequencies

‡`replace` is available only for `tabi`.

Options

Main

`chi2` calculates and displays Pearson's χ^2 for the hypothesis that the rows and columns in a two-way table are independent. `chi2` may not be specified if `aweights` or `iweights` are specified.

`exact[(#)]` displays the significance calculated by Fisher's exact test and may be applied to $r \times c$ as well as to 2×2 tables. For 2×2 tables, both one- and two-sided probabilities are displayed. For $r \times c$ tables, two-sided probabilities are displayed. The optional positive integer # is a multiplier on the amount of memory that the command is permitted to consume. The default is 1. This option should not be necessary for reasonable $r \times c$ tables. If the command terminates with error 910, try `exact(2)`. The maximum row or column dimension allowed when computing Fisher's exact test is the maximum row or column dimension for `tabulate` (see [R] **Limits**).

`gamma` displays Goodman and Kruskal's gamma along with its asymptotic standard error. `gamma` is appropriate only when both variables are ordinal. `gamma` may not be specified if `aweights` or `iweights` are specified.

`lrchi2` displays the likelihood-ratio χ^2 statistic. `lrchi2` may not be specified if `aweights` or `iweights` are specified.

`taub` displays Kendall's τ_b along with its asymptotic standard error. `taub` is appropriate only when both variables are ordinal. `taub` may not be specified if `aweights` or `iweights` are specified.

`V` (note capitalization) displays Cramér's V. `V` may not be specified if `aweights` or `iweights` are specified.

`cchi2` displays each cell's contribution to Pearson's χ^2 in a two-way table.

`column` displays the relative frequency of each cell within its column in a two-way table.

`row` displays the relative frequency of each cell within its row in a two-way table.

`clrchi2` displays each cell's contribution to the likelihood-ratio χ^2 in a two-way table.

`cell` displays the relative frequency of each cell in a two-way table.

`expected` displays the expected frequency of each cell in a two-way table.

nofreq suppresses the printing of the frequencies.

rowsort and colsort specify that the rows and columns, respectively, be presented in order of observed frequency.

By default, rows and columns are presented in ascending order of the row and column variable. For instance, if you type tabulate a b and a takes on the values 2, 3, and 5, then the first row of the table will correspond to a = 2; the second row will correspond to a = 3; and the third row will correspond to a = 5.

rowsort specifies that the rows instead be presented in descending order of observed frequency of the values. If you type twoway a b, rowsort, the most frequently observed value of a will be listed in the first row, the second most frequently observed value of a in the second row, and so on. If there are rows with equal frequencies, they will be presented in ascending order of the values of a. If a = 5 occurs with frequency 1,000 and values a = 2 and a = 3 each occur with frequency 500, the rows will be presented in the order a = 5, a = 2, and a = 3.

colsort does the same as rowsort, except with the columns and the column variable.

rowsort and colsort may be specified together.

missing requests that missing values be treated like other values in calculations of counts, percentages, and other statistics.

wrap requests that Stata take no action on wide, two-way tables to make them readable. Unless wrap is specified, wide tables are broken into pieces to enhance readability.

[no]key suppresses or forces the display of a key above two-way tables. The default is to display the key if more than one cell statistic is requested, and otherwise to omit it. key forces the display of the key. nokey suppresses its display.

nolabel causes the numeric codes to be displayed rather than the value labels.

nolog suppresses the display of the log for Fisher's exact test. Using Fisher's exact test requires counting all tables that have a probability exceeding that of the observed table given the observed row and column totals. The log counts down each stage of the network computations, starting from the number of columns and counting down to 1, displaying the number of nodes in the network at each stage. A log is not displayed for 2×2 tables.

firstonly, available only with tab2, restricts the output to only those tables that include the first variable in *varlist*. Use this option to interact one variable with a set of others.

Collect

`collect` and `collect(`[*cname*][, *collect_options*]`)` specify that results be posted to a collection. This collection produces a table that you can customize and publish to Microsoft Word, Microsoft Excel, PDF, HTML, LaTeX, SMCL, or Markdown. Output does not change when these options are specified. Use `collect preview` to see the customizeable table.

`collect` is a shortcut for `collect(Tabulate)`.

cname specifies that a collection named *cname* be associated with the collected results. The default is `Tabulate`.

`append` specifies that results be appended to collection *cname*.

`replace` permits `tabulate` to overwrite an existing collection. This option is implied for collection `Tabulate` when `append` is not specified.

`label(`*filename*`)` specifies the *filename* containing the collection labels to use for your table. Labels in *filename* will be loaded into the collection, and any labels not specified in *filename* will be taken from the labels defined in `c(collect_label)`. The default is to use only the collection labels set in `c(collect_label)`; see [TABLES] **set collect_label**.

`style(`*filename*[, `override`]`)` specifies the *filename* containing the collection styles to use for your table. The default collection styles will be discarded, and only the collection styles in *filename* will be applied.

If you prefer the default collection styles but also want to apply any styles in *filename*, specify `override`. If there are conflicts between the default collection styles and those in *filename*, the ones in *filename* will take precedence.

The default is to use only the collection styles set in `c(tabulate_style)`; see [TABLES] **set tabulate_style**.

Advanced

`matcell(`*matname*`)` saves the reported frequencies in *matname*. This option is for use by programmers.

`matrow(`*matname*`)` saves the numeric values of the $r \times 1$ row stub in *matname*. This option is for use by programmers. `matrow()` may not be specified if the row variable is a string.

`matcol(`*matname*`)` saves the numeric values of the $1 \times c$ column stub in *matname*. This option is for use by programmers. `matcol()` may not be specified if the column variable is a string.

`replace` indicates that the immediate data specified as arguments to the `tabi` command be left as the current data in place of whatever data were there.

The following option is available with `tabulate` but is not shown in the dialog box:

`all` is equivalent to specifying `chi2 lrchi2 V gamma taub`. Note the omission of `exact`. When `all` is specified, `no` may be placed in front of the other options. `all noV` requests all association measures except Cramér's V (and Fisher's exact). `all exact` requests all association measures, including Fisher's exact test. `all` may not be specified if `aweights` or `iweights` are specified.

Limits

Two-way tables may have a maximum of 1,200 rows and 80 columns (Stata/MP and Stata/SE) or 300 rows and 20 columns (Stata/BE). If larger tables are needed, see [R] **table**.

Remarks and examples

Remarks are presented under the following headings:

tabulate
Measures of association
N-way tables
Weighted data
Tables with immediate data
tab2
Publish your tables
Video examples

For each value of a specified variable (or a set of values for a pair of variables), `tabulate` reports the number of observations with that value. The number of times a value occurs is called its *frequency*.

tabulate

▷ Example 1

`tabulate` will make two-way tables if we specify two variables following the word `tabulate`. In our highway dataset, we have a variable called `rate` that divides the accident rate into three categories: below 4, 4–7, and above 7 per million vehicle miles. Let's make a table of the speed limit category and the accident-rate category:

```
. use https://www.stata-press.com/data/r19/hiway2
(Minnesota highway data, 1973)
. tabulate spdcat rate
```

Speed limit category	Accident rate per million vehicle miles			
	Below 4	4–7	Above 7	Total
40 to 50	3	5	3	11
55 to 50	19	6	1	26
Above 60	2	0	0	2
Total	24	11	4	39

The table indicates that three stretches of highway have an accident rate below 4 and a speed limit of 40 to 50 miles per hour. The table also shows the row and column sums (called the *marginals*). The number of highways with a speed limit of 40 to 50 miles per hour is 11, which is the same result we obtained in our previous one-way tabulations.

Stata can present this basic table in several ways—16, to be precise—and we will show just a few below. It might be easier to read the table if we included the row percentages. For instance, of 11 highways in the lowest speed limit category, three are also in the lowest accident-rate category. Three-elevenths amounts to some 27.3%. We can ask Stata to fill in this information for us by using the `row` option:

```
. tabulate spdcat rate, row
```

Key
frequency *row percentage*

Speed limit category	Accident rate per million vehicle miles: Below 4	4–7	Above 7	Total
40 to 50	3 27.27	5 45.45	3 27.27	11 100.00
55 to 50	19 73.08	6 23.08	1 3.85	26 100.00
Above 60	2 100.00	0 0.00	0 0.00	2 100.00
Total	24 61.54	11 28.21	4 10.26	39 100.00

The number listed below each frequency is the percentage of cases that each cell represents out of its row. That is easy to remember because we see 100% listed in the "`Total`" column. The bottom row is also informative. We see that 61.54% of all the highways in our dataset fall into the lowest accident-rate category, that 28.21% are in the middle category, and that 10.26% are in the highest.

tabulate can calculate column percentages and cell percentages, as well. It does so when we specify the column or cell options, respectively. We can even specify them together. Below is a table that includes everything:

```
. tabulate spdcat rate, row column cell
```

Key
frequency *row percentage* *column percentage* *cell percentage*

Speed limit category	Accident rate per million vehicle miles			
	Below 4	4–7	Above 7	Total
40 to 50	3	5	3	11
	27.27	45.45	27.27	100.00
	12.50	45.45	75.00	28.21
	7.69	12.82	7.69	28.21
55 to 50	19	6	1	26
	73.08	23.08	3.85	100.00
	79.17	54.55	25.00	66.67
	48.72	15.38	2.56	66.67
Above 60	2	0	0	2
	100.00	0.00	0.00	100.00
	8.33	0.00	0.00	5.13
	5.13	0.00	0.00	5.13
Total	24	11	4	39
	61.54	28.21	10.26	100.00
	100.00	100.00	100.00	100.00
	61.54	28.21	10.26	100.00

The number at the top of each cell is the frequency count. The second number is the row percentage—they sum to 100% going across the table. The third number is the column percentage—they sum to 100% going down the table. The bottom number is the cell percentage—they sum to 100% going down all the columns and across all the rows. For instance, highways with a speed limit above 60 miles per hour and in the lowest accident rate category account for 100% of highways with a speed limit above 60 miles per hour; 8.33% of highways in the lowest accident-rate category; and 5.13% of all our data.

A fourth option, nofreq, tells Stata not to print the frequency counts. To construct a table consisting of only row percentages, we type

```
. tabulate spdcat rate, row nofreq
```

Speed limit category	Accident rate per million vehicle miles: Below 4	4–7	Above 7	Total
40 to 50	27.27	45.45	27.27	100.00
55 to 50	73.08	23.08	3.85	100.00
Above 60	100.00	0.00	0.00	100.00
Total	61.54	28.21	10.26	100.00

Measures of association

▷ Example 2

tabulate will calculate the Pearson χ^2 test for the independence of the rows and columns if we specify the chi2 option. Suppose that we have 1980 census data on 956 cities in the United States and wish to compare the age distribution across regions of the country. Assume that agecat is the median age in each city and that region denotes the region of the country in which the city is located.

```
. use https://www.stata-press.com/data/r19/citytemp2
(City temperature data)
. tabulate region agecat, chi2
```

Census region	Age category: 19–29	30–34	35+	Total
NE	46	83	37	166
N Cntrl	162	92	30	284
South	139	68	43	250
West	160	73	23	256
Total	507	316	133	956

Pearson chi2(6) = 61.2877 Pr = 0.000

We obtain the standard two-way table and, at the bottom, a summary of the χ^2 test. Stata informs us that the χ^2 associated with this table has 6 degrees of freedom and is 61.29. The observed differences are significant.

The table is, perhaps, easier to understand if we suppress the frequencies and print just the row percentages:

```
. tabulate region agecat, row nofreq chi2
```

Census region	Age category 19–29	30–34	35+	Total
NE	27.71	50.00	22.29	100.00
N Cntrl	57.04	32.39	10.56	100.00
South	55.60	27.20	17.20	100.00
West	62.50	28.52	8.98	100.00
Total	53.03	33.05	13.91	100.00

Pearson chi2(6) = 61.2877 Pr = 0.000

◁

▷ Example 3

We have data on dose level and outcome for a set of patients and wish to evaluate the association between the two variables. We can obtain all the association measures by specifying the `all` and `exact` options:

```
. use https://www.stata-press.com/data/r19/dose
. tabulate dose function, all exact
Enumerating sample-space combinations:
stage 3:  enumerations = 1
stage 2:  enumerations = 9
stage 1:  enumerations = 0
```

Dosage	Function < 1 hr	1 to 4	4+	Total
1/day	20	10	2	32
2/day	16	12	4	32
3/day	10	16	6	32
Total	46	38	12	96

```
          Pearson chi2(4) =   6.7780   Pr = 0.148
 Likelihood-ratio chi2(4) =   6.9844   Pr = 0.137
               Cramér's V =   0.1879
                    gamma =   0.3689  ASE = 0.129
          Kendall's tau-b =   0.2378  ASE = 0.086
           Fisher's exact =                 0.145
```

We find evidence of association but not enough to be truly convincing.

If we had not also specified the `exact` option, we would not have obtained Fisher's exact test. Stata can calculate this statistic both for 2×2 tables and for $r \times c$. For 2×2 tables, the calculation is almost instant. On more general tables, however, the calculation can take longer.

We carefully constructed our example so that `all` would be meaningful. Kendall's τ_b and Goodman and Kruskal's gamma are relevant only when both dimensions of the table can be ordered, say, from low to high or from worst to best. The other statistics, however, are always applicable.

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❑ Technical note

Be careful when attempting to compute the p-value for Fisher's exact test because the number of tables that contribute to the p-value can be extremely large and a solution may not be feasible. The errors that are indicative of this situation are errors 910, exceeded memory limitations, and 1401, integer overflow due to large row-margin frequencies. If execution terminates because of memory limitations, use `exact(2)` to permit the algorithm to consume twice the memory, `exact(3)` for three times the memory, etc. The default memory usage should be sufficient for reasonable tables.

❑

N-way tables

If you need more than two-way tables, your best alternative to is use `table`, not `tabulate`; see [R] **table**.

The technical note below shows you how to use `tabulate` to create a sequence of two-way tables that together form, in effect, a three-way table, but using `table` is easy and produces prettier results:

```
. use https://www.stata-press.com/data/r19/birthcat
(City data)
. table (agecat birthcat) (region), nototals
```

	Census region			
	NE	N Cntrl	South	West
Age category				
19–29				
Birth-rate category				
29–136	11	23	11	11
137–195	31	97	65	46
196–529	4	38	59	91
30–34				
Birth-rate category				
29–136	34	27	10	8
137–195	48	58	45	42
196–529	1	3	12	21
35+				
Birth-rate category				
29–136	34	26	27	18
137–195	3	4	7	4
196–529			4	

❑ Technical note

We can make n-way tables by combining the by *varlist*: prefix with `tabulate`. Continuing with the dataset of 956 cities, say that we want to make a table of age category by birth-rate category by region of the country. The birth-rate category variable is named `birthcat` in our dataset. To make separate tables for each age category, we would type

```
. by agecat, sort: tabulate birthcat region
```

```
-> agecat = 19-29
```

Birth-rate category	Census region NE	N Cntrl	South	West	Total
29-136	11	23	11	11	56
137-195	31	97	65	46	239
196-529	4	38	59	91	192
Total	46	158	135	148	487

```
-> agecat = 30-34
```

Birth-rate category	Census region NE	N Cntrl	South	West	Total
29-136	34	27	10	8	79
137-195	48	58	45	42	193
196-529	1	3	12	21	37
Total	83	88	67	71	309

```
-> agecat = 35+
```

Birth-rate category	Census region NE	N Cntrl	South	West	Total
29-136	34	26	27	18	105
137-195	3	4	7	4	18
196-529	0	0	4	0	4
Total	37	30	38	22	127

❑

Weighted data

▷ Example 4

`tabulate` can process weighted as well as unweighted data. As with all Stata commands, we indicate the weight by specifying the [*weight*] modifier; see **[U] 11.1.6 weight**.

Continuing with our dataset of 956 cities, we also have a variable called `pop`, the population of each city. We can make a table of region by age category, weighted by population, by typing

```
. tabulate region agecat [fweight=pop]
```

Census region	Age category 19–29	30–34	35+	Total
NE	4721387	10421387	5323610	20466384
N Cntrl	16901550	8964756	4015593	29881899
South	13894254	7686531	4141863	25722648
West	16698276	7755255	2375118	26828649
Total	52215467	34827929	15856184	102899580

If we specify the `cell`, `column`, or `row` options, they will also be appropriately weighted. Below, we repeat the table, suppressing the counts and substituting row percentages:

```
. tabulate region agecat [fweight=pop], nofreq row
```

Census region	Age category 19–29	30–34	35+	Total
NE	23.07	50.92	26.01	100.00
N Cntrl	56.56	30.00	13.44	100.00
South	54.02	29.88	16.10	100.00
West	62.24	28.91	8.85	100.00
Total	50.74	33.85	15.41	100.00

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Tables with immediate data

▷ Example 5

`tabi` ignores the dataset in memory and uses as the table the values that we specify on the command line:

```
. tabi 30 18 \ 38 14
```

row	col 1	2	Total
1	30	18	48
2	38	14	52
Total	68	32	100

```
          Fisher's exact =                 0.289
  1-sided Fisher's exact =                 0.179
```

We may specify any of the options of `tabulate` and are not limited to 2×2 tables:

```
. tabi 30 18 38 \ 13 7 22, chi2 exact

Enumerating sample-space combinations:
stage 3:  enumerations = 1
stage 2:  enumerations = 3
stage 1:  enumerations = 0
```

row	col 1	2	3	Total
1	30	18	38	86
2	13	7	22	42
Total	43	25	60	128

```
          Pearson chi2(2) =   0.7967   Pr = 0.671
           Fisher's exact =                 0.707
. tabi 30 13 \ 18 7 \ 38 22, all exact col

+-----------------------+
| Key                   |
|-----------------------|
|     frequency         |
| column percentage     |
+-----------------------+

Enumerating sample-space combinations:
stage 3:  enumerations = 1
stage 2:  enumerations = 3
stage 1:  enumerations = 0
```

row	col 1	2	Total
1	30	13	43
	34.88	30.95	33.59
2	18	7	25
	20.93	16.67	19.53
3	38	22	60
	44.19	52.38	46.88
Total	86	42	128
	100.00	100.00	100.00

```
          Pearson chi2(2) =   0.7967   Pr = 0.671
 Likelihood-ratio chi2(2) =   0.7985   Pr = 0.671
              Cramér's V =    0.0789
                   gamma =    0.1204  ASE = 0.160
           Kendall's tau-b =   0.0630  ASE = 0.084
            Fisher's exact =                 0.707
```

For 2×2 tables, both one- and two-sided Fisher's exact probabilities are displayed; this is true of both `tabulate` and `tabi`. See *Cumulative incidence data* and *Case–control data* in [R] **Epitab** for more discussion on the relationship between one- and two-sided probabilities.

◁

❑ Technical note

tabi, as with all immediate commands, leaves any data in memory undisturbed. With the replace option, however, the data in memory are replaced by the data from the table:

```
. tabi 30 18 \ 38 14, replace
```

row	col 1	2	Total
1	30	18	48
2	38	14	52
Total	68	32	100

```
          Fisher's exact =                 0.289
  1-sided Fisher's exact =                 0.179
. list
```

	row	col	pop
1.	1	1	30
2.	1	2	18
3.	2	1	38
4.	2	2	14

With this dataset, you could re-create the above table by typing

```
. tabulate row col [fweight=pop], exact
```

row	col 1	2	Total
1	30	18	48
2	38	14	52
Total	68	32	100

```
          Fisher's exact =                 0.289
  1-sided Fisher's exact =                 0.179
```

❑

tab2

tab2 is a convenience tool. Typing

```
. tab2 myvar thisvar thatvar, chi2
```

is equivalent to typing

```
. tabulate myvar thisvar, chi2
. tabulate myvar thatvar, chi2
. tabulate thisvar thatvar, chi2
```

Publish your tables

With the collect option, tabulate and tabi post the tabulated values to a collection named Tabulate and set it as the current collection. tab2 appends the resulting tabulations into a single collection named Tabulate and sets it as the current collection. With collections, you can customize the look of your table, then publish it to HTML, Word, LaTeX, PDF, Excel, or another format appropriate for your report.

The default collect layout for tab2 when the firstonly option is not specified produces a table for the first pair of variables. You must specify the corresponding layout to get a table for the other two-way tabulations.

With the by prefix, tabulate and tab2 append the resulting tabulations into a single collection, and the default layout produces a separate table for each by group.

If you are not familiar with collections, see [TABLES] **Intro**. The predefined styles for tabulate are documented in [TABLES] **Predefined styles**.

▷ Example 6

Recall the example where we used the all option to show all the supported measures of association. Let's use tabulate with the collect option to produce a collection with its tabulated values.

```
. tabulate dose function, all exact collect
Enumerating sample-space combinations:
stage 3:  enumerations = 1
stage 2:  enumerations = 9
stage 1:  enumerations = 0
```

Dosage	Function < 1 hr	1 to 4	4+	Total
1/day	20	10	2	32
2/day	16	12	4	32
3/day	10	16	6	32
Total	46	38	12	96

```
          Pearson chi2(4) =   6.7780   Pr = 0.148
 Likelihood-ratio chi2(4) =   6.9844   Pr = 0.137
               Cramér's V =   0.1879
                    gamma =   0.3689  ASE = 0.129
          Kendall's tau-b =   0.2378  ASE = 0.086
           Fisher's exact =                 0.145
```

The output does not change; however, we can use the collect dir command to see that tabulate created a collection named Tabulate.

```
. collect dir
Collections in memory
Current: Tabulate
```

Name	No. items
Tabulate	28

In this collection, frequencies are tagged with `result[frequency]`. The tabulated variables (that is, `dose` and `function`) are added to the collection as dimensions and are used to tag the collected results. In addition to the names, labels, level values, and value labels of the tabulated variables, these dimensions each have the `__margCode__` level with the `Total` label for tagging the marginal frequencies. Here we use `collect label list` to show the levels and labels of each tabulated variable dimension.

```
. collect label list dose
  Collection: Tabulate
   Dimension: dose
       Label: Dosage
Level labels:
           1  1/day
           2  2/day
           3  3/day
__margCode__  Total

. collect label list function
  Collection: Tabulate
   Dimension: function
       Label: Function
Level labels:
           1  < 1 hr
           2  1 to 4
           3  4+
__margCode__  Total
```

When measures of association are requested, `tabulate` uses a style that defines composite results for them. Each measure of association has a corresponding composite result that composes it in a single cell of a second table.

Measure of association	Option	Composite result
Pearson's χ^2	`chi2`	`Pearson_ma`
Fisher's exact	`exact`	`Fisher_ma`
1-sided Fisher's exact	`exact`	`Fisher1_ma`
Likelihood ratio	`lr`	`LR_ma`
Cramér's V	`V`	`Cramer_ma`
Goodman and Kruskal's γ	`gamma`	`GoodmanKruskal_ma`
Kendall's τ_b	`taub`	`Pearson_ma`

tabulate constructs a default layout, so you can view your customizable table with the collect preview command. Here we use the collect layout command to report the default layout specification and corresponding table.

```
. collect layout

Collection: Tabulate
      Rows: dose#result result
   Columns: function _overall
    Tables: cmdset#item_type
   Table 1: 5 x 4
   Table 2: 6 x 1
```

	Function			
	< 1 hr	1 to 4	4+	Total
Dosage				
1/day	20	10	2	32
2/day	16	12	4	32
3/day	10	16	6	32
Total	46	38	12	96

```
          Pearson χ²(4) = 6.7780 Pr = 0.1481
               LR χ²(4) = 6.9844 Pr = 0.1367
                          Cramér's V = 0.1879
Goodman and Kruskal's γ = 0.3689 ASE = 0.1287
         Kendall's τ_b = 0.2378 ASE = 0.0865
                       Fisher's exact = 0.1449
```

We can make further changes to the table with the collect suite of commands. But we are happy with this layout and ready to publish the table to an HTML file with collect export. We simply specify the filename to which we want to export it.

```
. collect export tab2.html
(collection Tabulate exported to file tab2.html)
```

With collect export, you can publish the table to several formats, such as HTML, PDF, and LaTeX files, by specifying the appropriate file extension.

◁

Video examples

Pearson's chi-squared and Fisher's exact test in Stata

Tables and cross-tabulations in Stata

Cross-tabulations and chi-squared tests calculator

Stored results

`tabulate`, `tab2`, and `tabi` store the following in `r()`:

Scalars

`r(N)`	number of observations	`r(p_exact)`	Fisher's exact p
`r(r)`	number of rows	`r(chi2_lr)`	likelihood-ratio χ^2
`r(c)`	number of columns	`r(p_lr)`	p-value for likelihood-ratio test
`r(chi2)`	Pearson's χ^2 test	`r(CramersV)`	Cramér's V
`r(p)`	p-value for of Pearson's χ^2 test	`r(ase_gam)`	ASE of gamma
`r(gamma)`	gamma	`r(ase_taub)`	ASE of τ_b
`r(p1_exact)`	one-sided Fisher's exact p	`r(taub)`	τ_b

Macros

`r(collection)`	name of collection

`r(p1_exact)` is defined only for 2×2 tables. `r(collection)` is defined only when the `collect()` option is specified. Also, the `matrow()`, `matcol()`, and `matcell()` options allow you to obtain the row values, column values, and frequencies, respectively.

Methods and formulas

Let n_{ij}, $i = 1, \ldots, I$ and $j = 1, \ldots, J$, be the number of observations in the ith row and jth column. If the data are not weighted, n_{ij} is just a count. If the data are weighted, n_{ij} is the sum of the weights of all data corresponding to the (i, j) cell.

Define the row and column marginals as

$$n_{i\cdot} = \sum_{j=1}^{J} n_{ij} \qquad\qquad n_{\cdot j} = \sum_{i=1}^{I} n_{ij}$$

and let $n = \sum_i \sum_j n_{ij}$ be the overall sum. Also, define the concordance and discordance as

$$A_{ij} = \sum_{k>i} \sum_{l>j} n_{kl} + \sum_{k<i} \sum_{l<j} n_{kl} \qquad\qquad D_{ij} = \sum_{k>i} \sum_{l<j} n_{kl} + \sum_{k<i} \sum_{l>j} n_{kl}$$

along with twice the number of concordances $P = \sum_i \sum_j n_{ij} A_{ij}$ and twice the number of discordances $Q = \sum_i \sum_j n_{ij} D_{ij}$.

The Pearson χ^2 statistic with $(I-1)(J-1)$ degrees of freedom (so called because it is based on Pearson (1900); see Conover [1999, 240] and Fienberg [1980, 9]) is defined as

$$X^2 = \sum_i \sum_j \frac{(n_{ij} - m_{ij})^2}{m_{ij}}$$

where $m_{ij} = n_{i\cdot} n_{\cdot j} / n$.

The likelihood-ratio χ^2 statistic with $(I-1)(J-1)$ degrees of freedom (Fienberg 1980, 40) is defined as

$$G^2 = 2 \sum_i \sum_j n_{ij} \ln(n_{ij}/m_{ij})$$

Cramér's V (Cramér 1946) is a measure of association designed so that the attainable upper bound is 1. For 2×2 tables, $-1 \le V \le 1$, and otherwise, $0 \le V \le 1$.

$$V = \begin{cases} (n_{11}n_{22} - n_{12}n_{21}) / (n_{1\cdot} n_{2\cdot} n_{\cdot 1} n_{\cdot 2})^{1/2} & \text{for } 2 \times 2 \\ \{(X^2/n) / \min(I-1, J-1)\}^{1/2} & \text{otherwise} \end{cases}$$

Gamma (Goodman and Kruskal 1954, 1959, 1963, 1972; also see Agresti [2010,186–188]) ignores tied pairs and is based only on the number of concordant and discordant pairs of observations, $-1 \leq \gamma \leq 1$,

$$\gamma = (P - Q)/(P + Q)$$

with asymptotic variance

$$16 \sum_i \sum_j n_{ij}(QA_{ij} - PD_{ij})^2/(P + Q)^4$$

Kendall's τ_b (Kendall 1945; also see Agresti 2010, 188–189), $-1 \leq \tau_b \leq 1$, is similar to gamma, except that it uses a correction for ties,

$$\tau_b = (P - Q)/(w_r w_c)^{1/2}$$

with asymptotic variance

$$\frac{\sum_i \sum_j n_{ij}(2w_r w_c d_{ij} + \tau_b v_{ij})^2 - n^3 \tau_b^2 (w_r + w_c)^2}{(w_r w_c)^4}$$

where

$$\begin{aligned} w_r &= n^2 - \sum_i n_{i\cdot}^2 \\ w_c &= n^2 - \sum_j n_{\cdot j}^2 \\ d_{ij} &= A_{ij} - D_{ij} \\ v_{ij} &= n_{i\cdot} w_c + n_{\cdot j} w_r \end{aligned}$$

Fisher's exact test (Fisher 1935; Finney 1948; see Zelterman and Louis [1992, 293–301] for the 2×2 case) yields the probability of observing a table that gives at least as much evidence of association as the one actually observed under the assumption of no association. Holding row and column marginals fixed, the hypergeometric probability P of every possible table A is computed, and the

$$P = \sum_{T \in A} \Pr(T)$$

where A is the set of all tables with the same marginals as the observed table, T^*, such that $\Pr(T) \leq \Pr(T^*)$. For 2×2 tables, the one-sided probability is calculated by further restricting A to tables in the same tail as T^*. The first algorithm extending this calculation to $r \times c$ tables was Pagano and Halvorsen (1981); the one implemented here is the FEXACT algorithm by Mehta and Patel (1986). This is a search-tree clipping method originally published by Mehta and Patel (1983) with further refinements by Joe (1988) and Clarkson, Fan, and Joe (1993). Fisher's exact test is a permutation test. For more information on permutation tests, see Good (2005 and 2006) and Pesarin (2001).

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Also see

[R] **Epitab** — Tables for epidemiologists

[R] **table** — Table of frequencies, summaries, and command results

[R] **table twoway** — Two-way tabulation

[R] **tabstat** — Compact table of summary statistics

[R] **tabulate oneway** — One-way table of frequencies

[R] **tabulate, summarize()** — One- and two-way tables of summary statistics

[D] **collapse** — Make dataset of summary statistics

[SVY] **svy: tabulate oneway** — One-way tables for survey data

[SVY] **svy: tabulate twoway** — Two-way tables for survey data

[TABLES] **Intro** — Introduction

[XT] **xttab** — Tabulate xt data

[U] 12.6.3 Value labels

[U] 26 Working with categorical data and factor variables

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